Name ____ Date ___________ Period _____

Comparing All Seven Parent Functions

Algebra II | Unit 1 | Day 8 | TEK A2.2.A

Objective: I can compare the key attributes of all seven parent functions, name a function from its attributes alone, and find a relative maximum or minimum on a given interval.

Vocabulary

Relative maximum — the _____________ y-value on a given interval.

Relative minimum — the _____________ y-value on a given interval.

A function can have no overall max or min but still have one ____________________.

Part 1 — The Master Comparison Table

Fill this in from your Day 4-7 notes. This is your reference sheet for the rest of the year.

Function	Domain	Range	Symmetry	Asymptotes
$\sqrt{x}$	__________	__________	__________	__________
$\lvert x \rvert$	__________	__________	__________	__________
x^3	__________	__________	__________	__________
$\sqrt[3]{x}$	__________	__________	__________	__________
$1/x$	__________	__________	__________	__________
b^x	__________	__________	__________	__________
log base b	__________	__________	__________	__________

Part 2 — Intercepts and Extremes

Function	Intercept(s)	Minimum?	Maximum?
$\sqrt{x}$	____________	____________	__________
$\lvert x \rvert$	____________	____________	__________
x^3 and $\sqrt[3]{x}$	____________	____________	__________
$1/x$	____________	____________	__________
b^x	____________	____________	__________
log base b	____________	____________	__________

Only ________ of the seven have a minimum, and ________ has a maximum. Only ________ has no intercept at all.

Part 3 — Name That Function

Use only the attributes listed. Name the function and sketch its shape in the box.

1. Domain and range both all reals except 0; two branches; asymptotes at $x = 0$ and $y = 0$.
   Function: _______________________

2. Domain all reals, range $[0, \infty)$, folds across the y-axis, minimum of 0.
   Function: _______________________

3. Domain $(0, \infty)$, range all reals, passes through $(1, 0)$.
   Function: _______________________

4. Domain and range both $[0, \infty)$, no symmetry, no asymptotes.

Function: ______________________

Part 4 — Relative Max and Min on an Interval

A graph rises and falls between y = 1.5 and y = -1.5 over the interval from 0° to 360°.

Relative maximum on that interval: ___________________ Relative minimum: ___________________

Now consider $f(x) = x^3$ on the interval [-2, 2].

Relative maximum: _______________________ Relative minimum: _______________________

x^3 has no overall max or min, but on a ___________________ it has both. Why? _________________________

Part 5 — Sort by One Attribute

Which parent functions have a restricted domain? _____________________________

Which have any asymptote? _____________________________

Which have range all real numbers? _____________________________

Practice

Use your comparison tables. Show reasoning where there is room.

1. Two parent functions pass through (1, 1) and (0, 0). Name them. ___________________________
2. Which parent function never touches either axis? _______________________
3. Which two are reflections of each other across y = x? Give two different pairs.

   ____
4. A graph has a minimum but no maximum, and no asymptotes. Name both possibilities.

   ____
5. $f(x) = |x|$ on the interval [-3, 1]: minimum ____________ maximum ____________
6. Explain why a function can have a relative maximum on an interval but none overall.

   ____

   ____

Exit Ticket

Two parent functions have domain and range that are both all real numbers, and both have 180° rotational symmetry.

Name them: _________________________ Name one function with a restricted range: _________________________